

Fake Certificate Recognition

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B.Tech CSE(AI&DS)

Introduction/Problem Statement

- The rise in forged and manipulated certificates has become a major concern for institutions.
- Manual verification methods are time-consuming and often lack accuracy.
- There is a need for an automated system that can reliably analyze certificate data.
- This project aims to provide a fast, efficient, and accurate solution using OCR-based verification.

Objectives of the Project

- To automatically extract text from certificates using OCR technology.
- To identify signs of tampering or inconsistencies in the extracted certificate data.
- To provide a fast and accurate method for verifying the authenticity of certificates.
- To reduce manual effort and human errors in certificate validation.
- To deliver a simple and user-friendly system for uploading and checking certificates.

System Analysis:

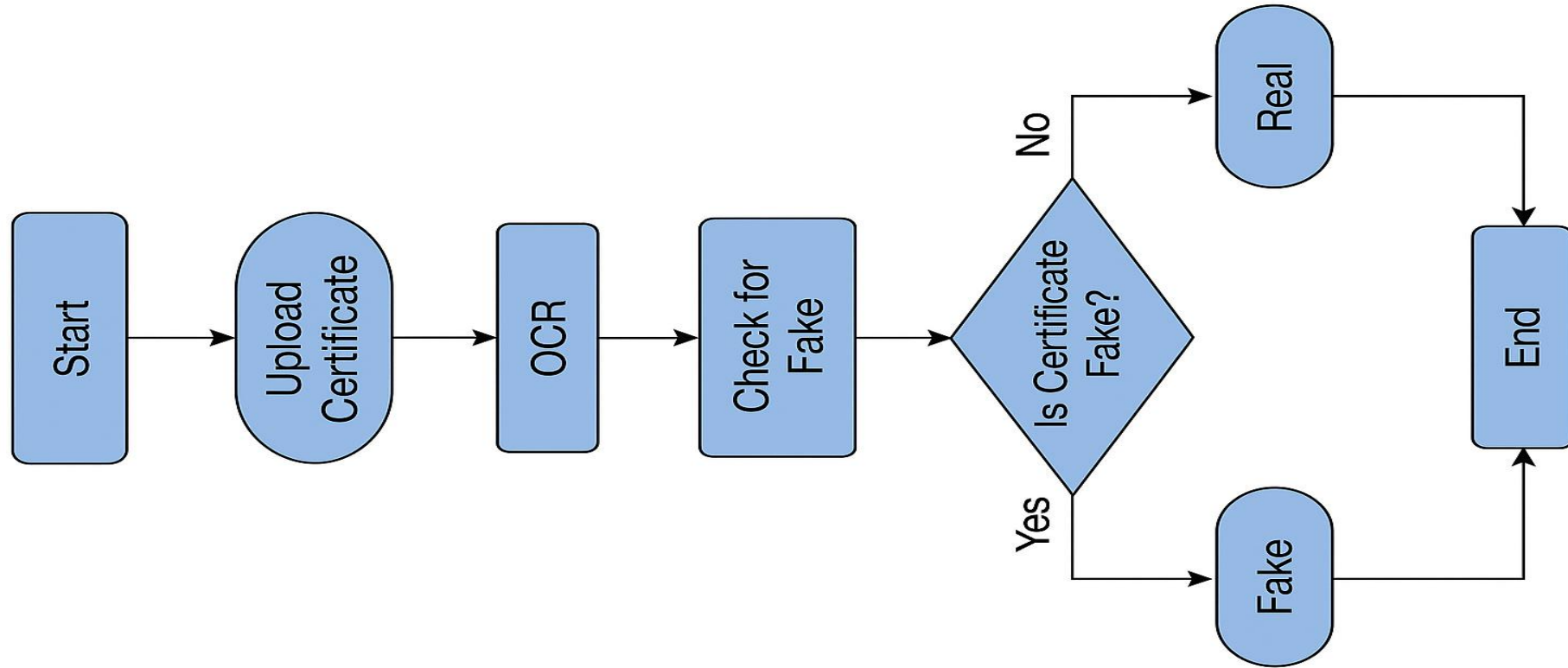
- Existing System:

1. The current certificate verification process is mostly manual, slow, and prone to human error.
2. There is no automated method to detect tampering, missing details, or fake formatting.
3. Verifiers rely on visual inspection, which is not always reliable or consistent..

- Proposed System:

1. The proposed system uses OCR to automatically extract and analyze certificate text.
2. It checks for inconsistencies, missing fields, and signs of fraud in real time.
3. The system provides faster, more accurate, and reliable verification with minimal manual work.

Data Flow Diagram (DFD) :



Certificate Upload Module

- Allows the user to upload a certificate image (JPG/PNG/PDF).
- Sends the uploaded file to the backend for processing.
- Acts as the starting point of the verification workflow.

2. OCR Extraction Module (Tesseract + Tess4J)

- Extracts text from the uploaded certificate image.
- Uses Tesseract OCR through Tess4J library on the backend.
- Converts image content into machine-readable text for further analysis.

3. Fake Certificate Detection Module

- Analyzes extracted text using simple rule-based checks.
- Verifies presence of keywords (e.g., “Name”, “Registration Number”, “University”).
- Detects irregularities like missing fields, spelling mistakes, or low text quality.
- Generates a “Real” or “Fake” decision.

4. Result Generation Module

- Combines extracted text + warnings + final verdict.
- Sends back a clean JSON response to the front end.
- Ensures user gets a readable and clear output.

5. Frontend Display Module

- A simple HTML page that accepts file upload and shows results.
- Displays extracted text, detected issues, and the final verification status.
- Ensures user-friendly interaction without needing any frameworks.

Implementation Details:

• Technologies Used

- **Backend:** Java, Spring Boot, Tess4J, Tesseract OCR
- **Frontend:** HTML, CSS, JavaScript
- **Build Tool:** Maven
- **IDE:** IntelliJ IDEA / VS Code
- **Server:** Embedded Tomcat (Spring Boot default)

Output:

Fake Certificate Recognition Service

Upload a certificate image to analyze it for common signs of a fake document using Tesseract OCR.

WhatsApp I...a83309e.jpg

Analysis Result:

```
{
  "warnings": [
    "Not enough expected fields found (only found 0).",
  ],
  "verdict": "Likely Fake",
  "extractedText": "fac ac DY UNIVERSITY\n<) JOY UNIVERSITY TIDCO\nana TANSAM A
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Advanced 3D MODELING & Simulation USING NX CAD\net ty Contact\n. 2 Hannah
Robert\n| a ES ACCOMMODATION AND FOOD ARE ALSO AVAILABLE @ +91 9384877934\nATA
VERY AFFORDABLE COST\nhttps://forms.gle/ja2Qf8U7iirDhQty6\n"
}
```

Fake Certificate Recognition Service

Upload a certificate image to analyze it for common signs of a fake document using Tesseract OCR.

10th_Marks...FFEBYD.jpg

Analysis Result:

```
{
  "warnings": [],
  "verdict": "Likely Genuine",
  "extractedText": "4 Board of Secondary Coucation ,\nTRADE ee\n'i yy uarioh ANDHRA PRADESH, INDIA pee Fr\nki Pea ul & er | Pa\nRest ce SECONDARY SCHOOL CERTIFICATE We NG:\n+
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Completed:

- Backend setup with Java + Spring Boot
- Tesseract OCR integration using Tess4J
- Extraction and fake-check logic
- Simple frontend upload page
- Working API to generate results

Ongoing / To be done (if needed):

- Adding more advanced fake detection rules
- UI design improvements

- **Result**

- Text successfully extracted from certificate images.
- Basic fake-detection checks working correctly.
- System gives clear output with warnings and final status.
- Frontend upload and result display working smoothly.

- **Discussion**

- OCR is effective for simple certificate verification.
- Accuracy depends on image quality.
- Current system works for basic validation.
- Can be improved with more detection rules or ML in the future.

A large, irregular splash of teal and light blue watercolor paint serves as the background for the text.

Thank You